OXYCODONE

High-alert drug
Controlled drug

CHEMICAL NAME 4,5α-epoxy-14-hydroxy-3-methoxy-17-methylmorphinan-6-one (as hydrochloride)
CLASS Strong opioid agonist (semi-synthetic)
TRADE NAMES OxyContin · OxyNorm · Targin (with naloxone) · Percocet (with paracetamol)

THERAPEUTIC USES / INDICATIONS

- Moderate to severe pain: post-operative, cancer pain.
- Chronic severe pain requiring continuous opioid therapy.
- Alternative when morphine is not tolerated.

ADMINISTRATION

- **PO:** immediate-release capsules/liquid or modified-release tablets (swallow whole — never crush or chew).
- **IV/SC** injection or infusion in hospital.
- Controlled-drug rules apply.

DOSAGES

- **Adult IR (opioid-naïve):** 5 mg every 4–6 h, titrated.
- **MR tablets:** 10 mg every 12 h, adjusted to response.
- **IV:** 1–10 mg slowly every 4 h.
- Oral oxycodone is ~1.5–2× as potent as oral morphine.
- Reduce dose in elderly, hepatic and renal impairment.

PHARMACOKINETICS

- **Onset:** PO 15–30 min. **Peak:** 1 h (IR). **Duration:** 4–6 h (IR); 12 h (MR).
- Oral bioavailability 60–87% (higher than morphine).
- Metabolised in liver (CYP3A4, CYP2D6). Excreted in urine. **t½:** 3–5 h.

PHARMACODYNAMICS

- Full **μ-opioid** agonist → analgesia, euphoria, sedation, respiratory depression.
- Antitussive and constipating effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Constipation, nausea, vomiting.
- Drowsiness, dizziness, itching.
- Dry mouth, sweating.

ADVERSE EFFECTS (SERIOUS)

- Respiratory depression, especially with sedatives.
- Hypotension; severe sedation.
- Dependence, misuse; withdrawal syndrome.
- Fatal overdose if MR tablets are crushed.

CONTRAINDICATIONS & CAUTIONS

- Acute respiratory depression; severe COPD or acute asthma.
- Paralytic ileus.
- Moderate–severe hepatic impairment (MR forms).
- **Caution:** head injury, elderly, renal impairment, hypothyroidism.

KEY INTERACTIONS

- **Benzodiazepines, alcohol, gabapentinoids:** respiratory depression.
- **CYP3A4 inhibitors** (clarithromycin, azole antifungals): ↑ levels.
- **MAOIs:** avoid.

NURSING MANAGEMENT

- Assess pain, RR, SpO_2 and sedation regularly.
- Keep naloxone available.
- Prescribe and give regular laxatives.
- Check opioid history and conversions carefully.
- Store and document as a controlled drug.

PATIENT EDUCATION

- Swallow MR tablets whole.
- Avoid alcohol and other sedatives.
- Do not drive if drowsy.
- Store safely; never share.

KEY POINTS

- Strong opioid; more potent orally than morphine.
- Watch RR and sedation; naloxone reverses.
- Never crush MR tablets.
- Give regular laxatives.

SOURCES CHECKED **[45]** Oxycodone Hydrochloride 1 mg/ml oral solution (UK SmPC) **[46]** Oxycodone Hydrochloride 10 mg/ml solution for injection or infusion (UK SmPC) **[47]** Oxycodone Hydrochloride, oral (US label) **[48]** Oxycodone Hydrochloride, oral (US label, Amneal Pharmaceuticals LLC)
Full references p. 432–457